Project Report Summary

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Summary Statistics of Credit Score & Balance

- Total number of customers – **SELECT COUNT(*) AS total_customers ;**
- Maximum Credit Score – **SELECT MAX("CreditScore") AS max_credit_score FROM customer_churn ;**
- Minimum Credit Score – **SELECT MIN("CreditScore") AS min_credit_score FROM customer_churn ;**
- Average Credit Score – **SELECT AVG("Credit Score") AS mean_credit_score FROM customer_churn;**
- Maximum Balance – **SELECT MAX("Balance") AS max_balance FROM customer_churn ;**
- Minimum Balance – **SELECT MIN("Balance") AS min_balance FROM customer_churn ;**
- Average Balance – **SELECT AVG("Balance") AS mean_balance FROM customer_churn;**

Mean & Median

- Mean of CreditScore = 650.53 and Median of CreditScore = 652
- Mean of Balance = 76485.89 and Median of Balance = 97208.46
- After calculating Mean and Median of both “Credit Score” and “Balance” is that For Credit Score, they are almost the same, so the Average is reliable. However, for Balance, the Median is a better indicator because we have a large number of ‘0’ balance customers who are outliers. They pull the Average down and make our customer base look less wealthy than it actually is."

Finding Mean & Median of Balance

```sql
-- Statistical Summary (Finding the mean and median of credit score and balance) :-

--Finding the average Average Balance:

SELECT
ROUND(AVG("Balance"),2) AS avg_balance
FROM customer_churn;

--Finding the median(middle value) of Balance:

SELECT "Balance" AS median_balance
FROM customer_churn
ORDER BY "Balance"
LIMIT 1 OFFSET (SELECT COUNT(*) FROM customer_churn) / 2;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	median_balance numeric
1	97208.46

Finding Mean & Median of Credit Score

```sql
--Finding the average Credit score:

SELECT
ROUND(AVG("CreditScore"),2) AS avg_credit_score
FROM customer_churn;

--Finding the median(middle value) of Credit score:

SELECT "CreditScore" AS median_credit_score
FROM customer_churn
ORDER BY "CreditScore"
LIMIT 1 OFFSET (SELECT COUNT(*) FROM customer_churn) / 2;
```

Data Output Messages Notifications

	avg_credit_score numeric
1	650.53

Using DDL and DML commands

- To explore the various segments of customers I have used DDL and DML commands to make different tables and fire different DML queries which are responsible for finding out different metrics.

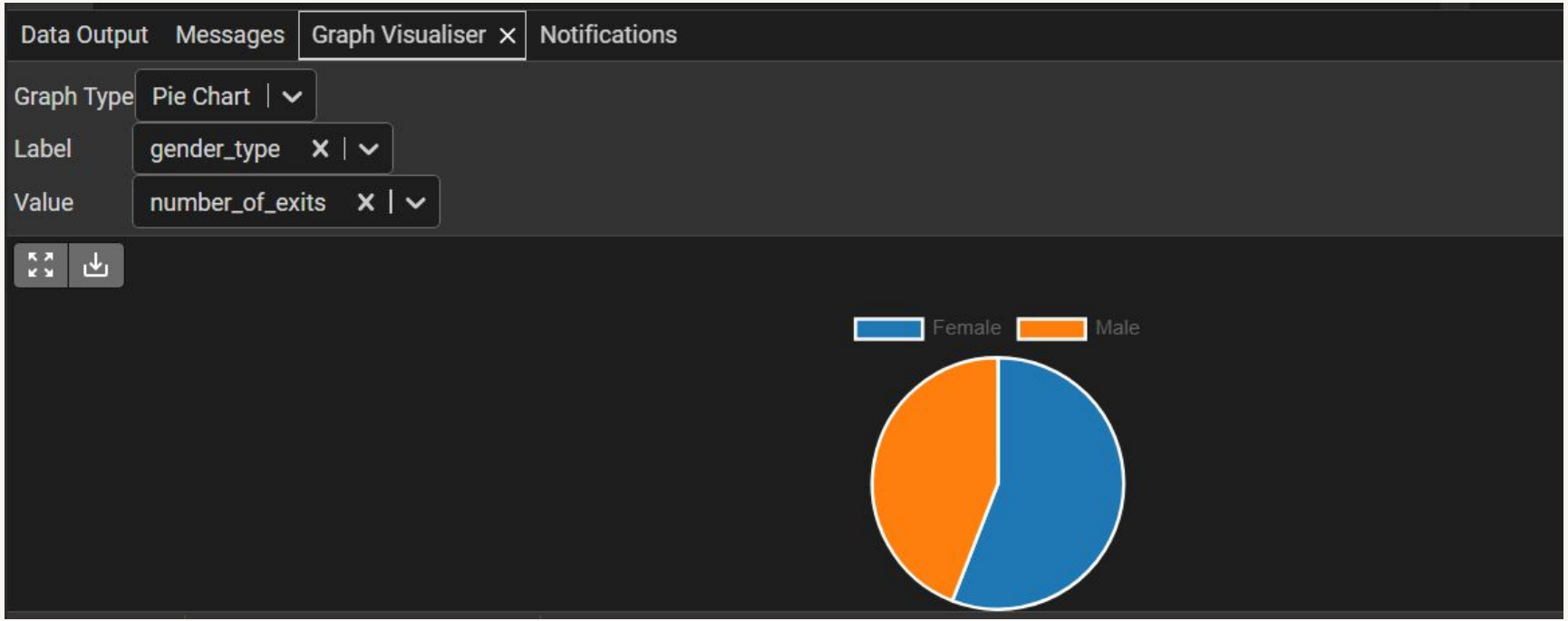
- In DDL queries, I have used **“CREATE”** Table command to create different tables and **“DROP”** command to drop the existing table when needed.

- In DML queries, I have used **“INSERT”** command to insert the values in different new tables and **“SELECT”** command to find out the key values and records from the newly created table. I have also used few aggregate functions to find the precised values for insights.

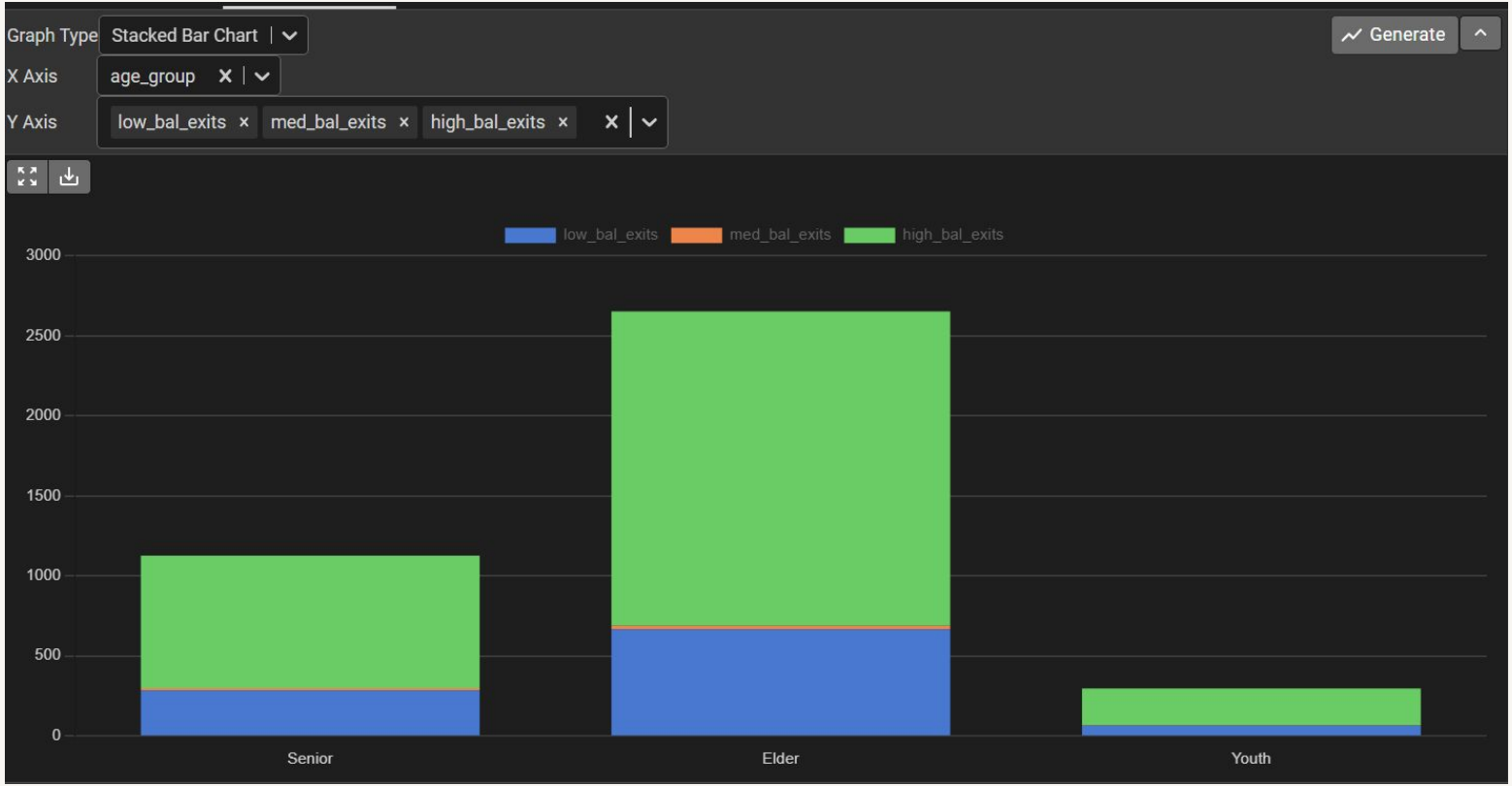
Top 5 Key Metrics responsible for customer churn

- Metric 1 - In this metric, it is clearly visible that **Females (2278)** exits more than **Male (1796)**.
- Metric 2 - In metric 2, Elder category people (31-50) years of age who have high balance (50000+) are exiting most. Even high balanced Seniors(50+) and Youth (>30) are also exiting more than low balanced and medium balanced people. Elder being the most **(2652)**.
- Metric 3 - People who uses only 1 Product have the most churned number in both high salary and low salary category **(1409)**. As the product services quantity increases, lesser is the churn numbers.
- Metric 4 - **Germany** has the highest exits **(814)** followed by **France (810). Spain** has most loyal customers.
- Metric 5 - People with **"Fair"** credit score **(500-650)** has the most churn numbers, whereas **"Poor"** credit score people are staying more.

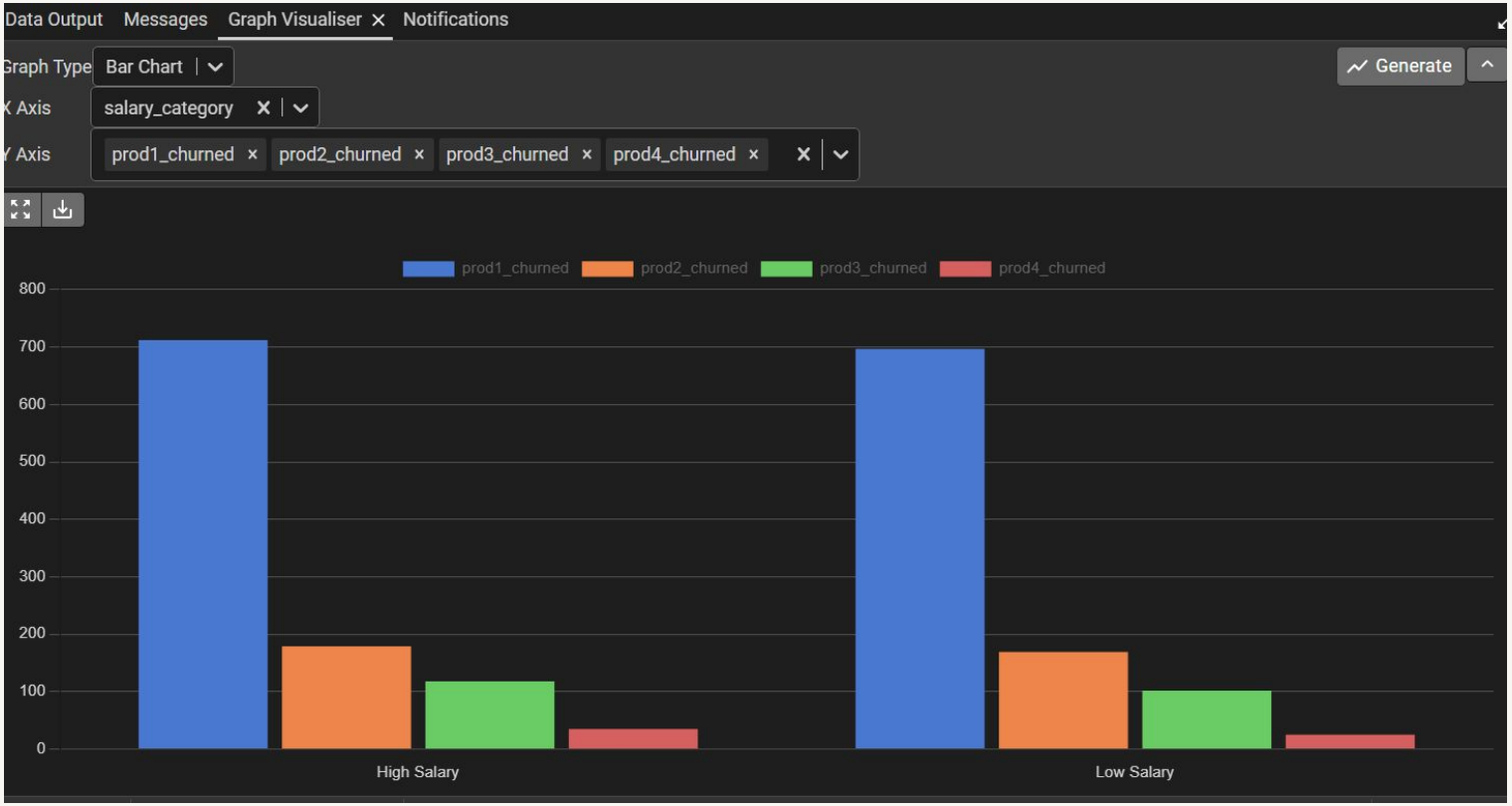
Metric 1 - Churn rate of different Gender



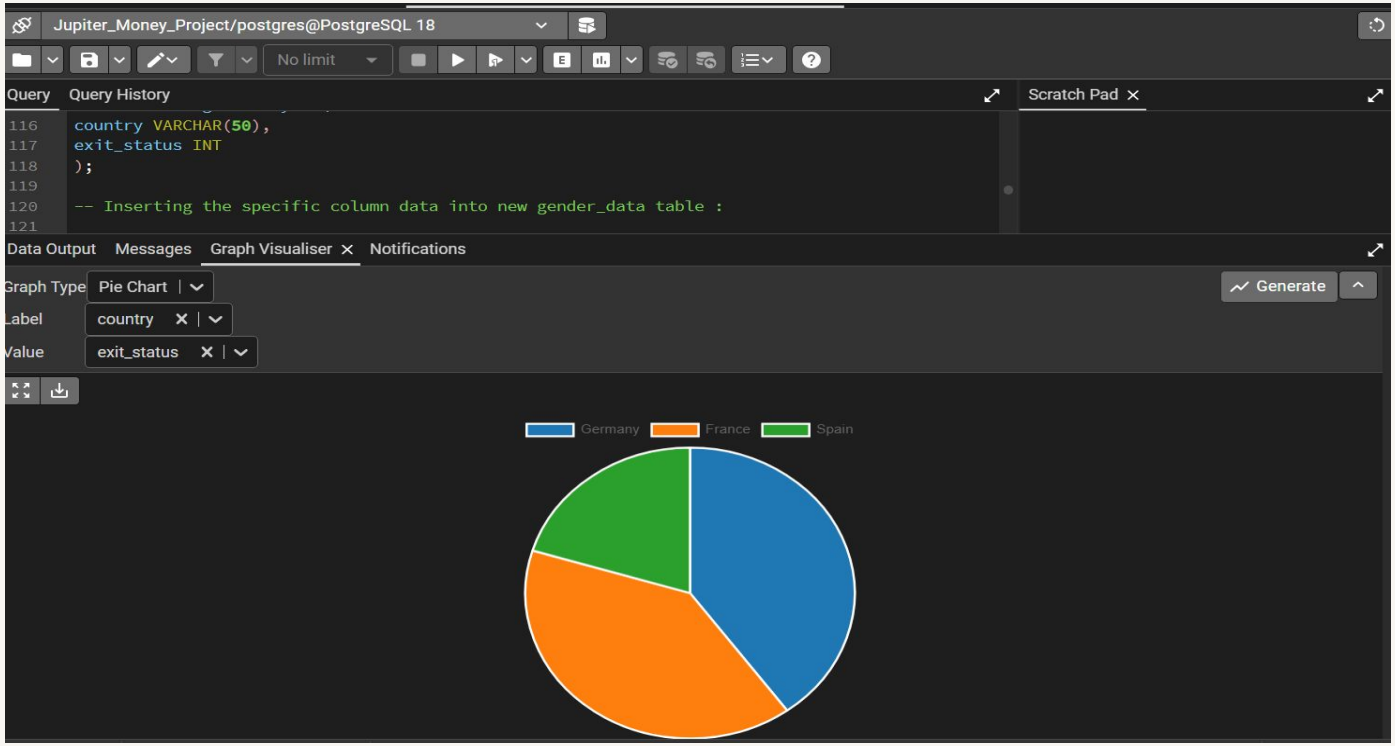
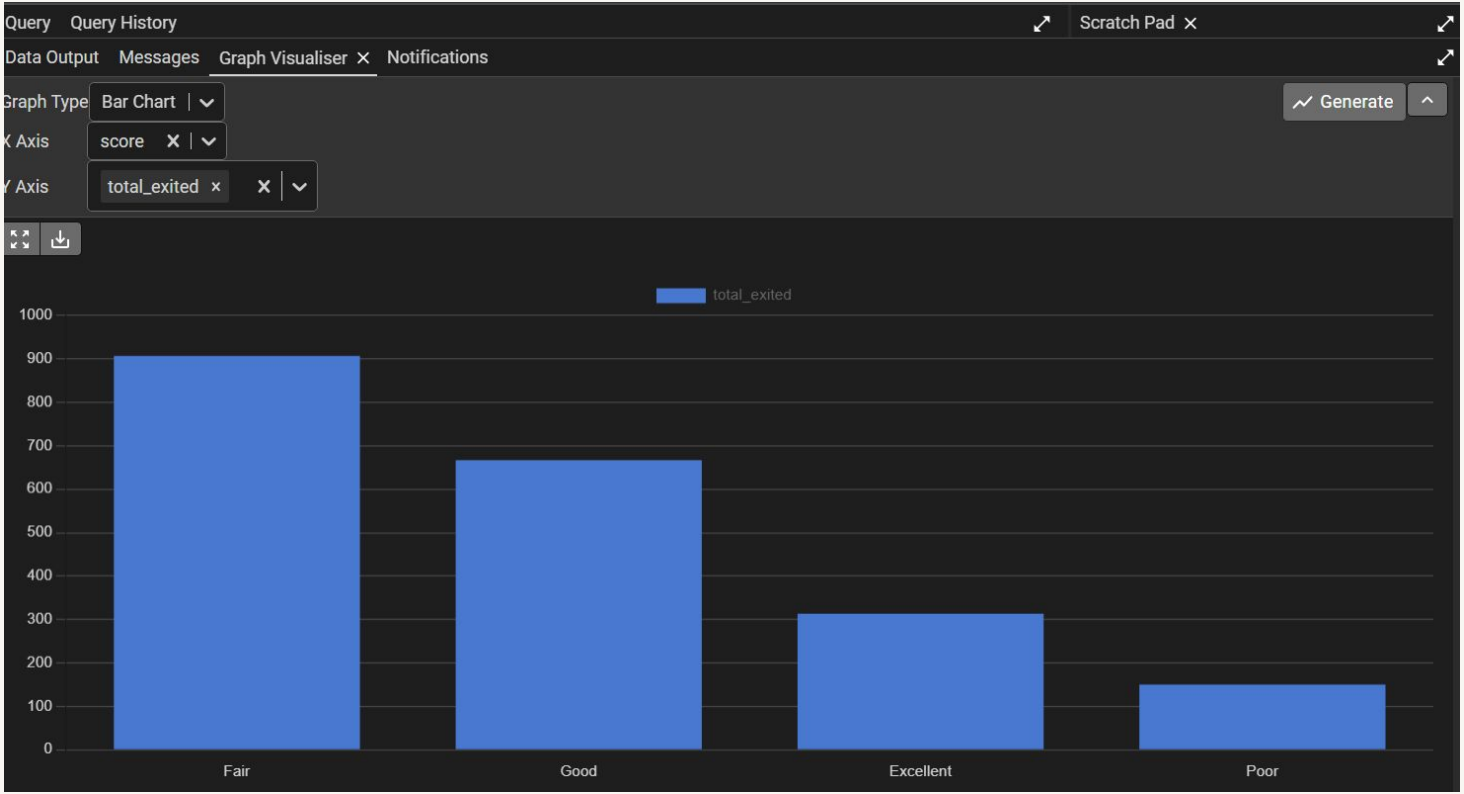
Metric 2 - Age Group and Balance Analysis



Metric 3 - Salary and Product Vs Churn Rate



Metric 4 - Churn rate based on Geography



Metric 5 - Visulization based on Credit Score